higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

11 December 2025

This marking guideline consists of 6 pages.

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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
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SECTION A

QUESTION 1: FLOW MEASUREMENT

1.1 1.1.1 It consists of a steel plate with a segment cut out.✓ The solid segment is at the top of the orifice plate with the open part at the bottom and flush with the pipe.✓ (1 × 2) (2)

1.1.2 Since the open segment is flush with the pipe, no solid material builds up on the upstream side of the orifice. (1)

1.1.3
- Venturi tube
- Flow nozzle
- Dall tube (3)

1.2 A weir is an obstruction in an open channel which constricts the flow✓ and causes it to fall over a crest.✓ Weirs consist of vertical plates (or concrete walls) with sharp crests across a flow.✓ The top of the plate can be straight or notched. (1 × 2) (2)

1.3 1.3.1
- Rate meters measure the velocity of flow.✓ The total quantity is calculated as the product of velocity and time.✓
- Volumetric flow meters close of quantity of fluid into compartments of known volume.✓ The total number of revolutions is multiplied by the compartment volume.✓

(2 × 2) (4)

1.3.2
- The primary element is defined as device which converts the measured quantity into a variable that is proportional to the measured quantity.
- The secondary element is a device the measures the variable from the primary element. (2)

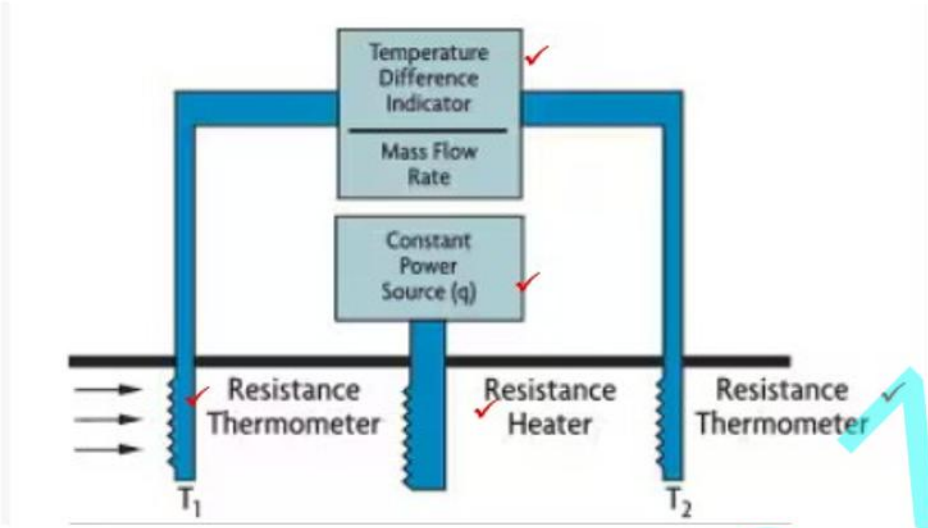
1.3.3 The purpose of the primary measuring flow element is to create a difference in pressure in a flowing fluid. (1)

[15]

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QUESTION 2

2.1



(5)

2.2 In the direct-heat version, a fixed amount of heat (q) is added by an electric heater.✓ As the process fluid flows through the pipe, heat is drawn and resistance temperature detectors (RTDs) measure the temperature rise✓ while the amount of electric heat introduced is held constant.✓ Once the fluid starts to circulate in the measuring tube, the heated temperature sensor is cooled by the fluid's motion, with the speed of the flow determining the degree of cooling.✓ As a result, the electric current needed to keep the temperature difference constant is a direct indication of mass flow.✓

The mass flow (m) is calculated on the basis of the measured temperature difference (T2-T1),✓ the meter coefficient (K), the electric heat rate (Q), and the specific heat of the fluid (Cp), as follows: $m = KQ/(Cp(T2-T1))$✓. (7)

2.3
- Temperature sensors and the heater protrude into the fluid stream which makes them vulnerable to corrosion and erosion damages.
- Pipe leakages are also possible due to these protrusions. (2)

2.4 Heated tube flow meters were developed to protect the heater and sensor elements from corrosion and any coating effects of the process. (1)

[15]

TOTAL SECTION A: 30

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SECTION B

QUESTION 3

3.1 3.1.1

- In communications to control frequencies
- To measure temperature
- To measure thickness of metal films
- To generate ultrasonic waves
- To measure moisture (5)

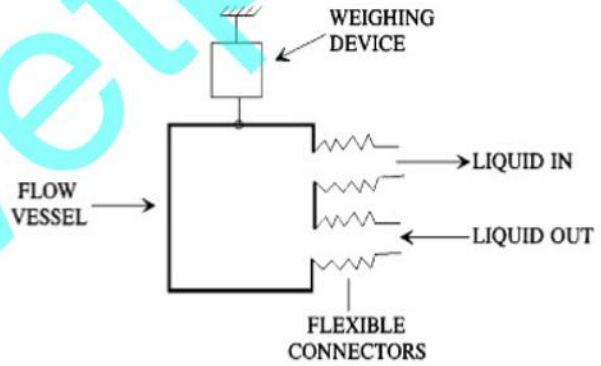
3.1.2

- For moisture measurement, the quartz crystals coated with hygroscopic material are exposed to the sample.✓
- Water from the sample is absorbed by the crystal coating ,thus increasing the total mass✓ and decreasing the oscillating frequency of the crystal.✓
- Two crystals are used,✓one crystal exposed to wet sample and the other to a dry reference gas.✓
- The frequency difference between the two crystals is in proportion to their mass difference✓ and the moisture content of the gas.✓ (7)

3.2

- The float viscometer must be installled vertically with the outlet at the top.
- The instrument should be carefully controlled.
- Process fluid temperature should be carefully controlled.
- The flow of the fluid should be smooth.
- The sample liquid should be void of any air or vapour entrainment.
- Install by-pass line.
- Make certain that the oerating conditiond are not exceeded.
- Recalibration should be made on a regular basis. (Any 5 × 1) (5)

3.3 3.3.1



(1 mark for each correct label) (5)

3.3.2 A fixed – volume vessel is used✓ through which the measured liquid flows continuously.✓ This vessel is weighed automatically by a scale, spring balance or pneumatic force-balanced transmitter.✓ Since the weight of the definite volume of the liquid is known,✓ the instrument can be calibrated to read directly in specific gravity units.✓ (5)

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3.4 3.4.1 Vibrating-reed viscometer (1)

3.4.2 When the drive coil is excited, it produces a pulsating magnetic field✓ which causes the drive armature to vibrate at the same frequency.✓ Mechanical vibration of the drive armature is transmitted to the probe.✓ The resistance to the shearing action caused by the probe vibration increases✓ with an increase in the process media viscosity.✓ (5)

[33]

QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1 To complete the electrical circuit with the glass membrane electrode.✓ It must provide a stable potential that is relatively invariable,✓ despite changes in either the chemical composition of the process stream. (1 × 2) (2)

4.2 4.2.1 Anode
4.2.2 Cathode
4.2.2 Cl^- = chlorine ions
4.2.2 H^+ = hydrogen ions
(4 × 1) (4)

4.3 Electrolysis is a process whereby certain substances undergo chemical change✓ (electrolytes) resulting in the split up into its ions,✓ thereby conducting electricity.✓ (1 × 3) (3)

4.4 When electrical current is passed through an electrolyte,✓ the electrodes acquire a certain surface potential✓ due to the loss or gain of electrons,✓ forming a simple voltaic cell. The electrodes are said to be polarized. (1 × 3) (3)

4.4 The pH of a solution is a measure of the acidity or alkalinity of the solution.✓✓ (1 × 2) (2)

[14]

QUESTION 5: AUTOMATIC CONTROL

5.1
- A measurement element
- An error detection element
- A final control element (3)

5.2
- ON/OFF sensing can be used for level sensing
- Positioning sensing
- Limit sensing (3)

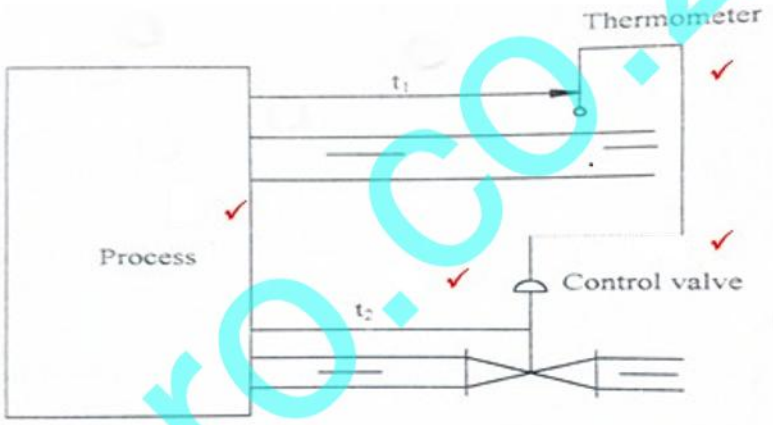
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5.3 5.3.1
- Temperature
- Flow
- Level
- Pressure (4)

5.3.2
- On/off or two step control mode
- Proportional control mode
- Proportional plus integral control mode
- Proportional plus derivative control mode
- Proportional plus integral plus derivative control mode (5)

5.4 5.4.1 The time interval between an alteration in a value of a signal✓ and its manifestation unchanged at a later part of the system.✓
(1 × 2) (2)

5.4.2



(4)

5.4.3
- Pipe diameters
- Length of the pipe (2)

[23]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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